GATE & PSU Master Question Bank

ARRAYS, FUNCTIONS, AND RECURSION

Comprehensive practice resource compiled from authentic exam blueprints over the last 10 years

SECTION A: RECURSION & FUNCTION TRACING

Question 1 (GATE CS 2026)

Consider the following recursive C function:

```
int dynamic_trace(int n, int m) {
    if (n <= 0 || m <= 0) return 0;
    if (n == m) return n + dynamic_trace(n - 1, m - 1);
    if (n > m) return n + dynamic_trace(n - m, m);
    return m + dynamic_trace(n, m - n);
}
```

What value is returned by the call ***dynamic_trace(7, 3)***?

Question 2 (GATE CS 2023)

Consider the following ANSI C program fragment:

```
int fun(int n) {
    static int count = 0;
    if (n <= 0) return count;
    count += n;
    return fun(n - 2) + fun(n - 1);
}
int main() {
    printf("%d", fun(4));
    return 0;
}
```

Determine the final value printed by the execution of the program.

Question 3 (GATE CS 2021)

Consider the following ANSI C program:

```
int foo(int x, int q) {
    if (x <= 1) return 1;
    if (x % q == 0) return foo(x / q, q);
    else return foo(x - 1, q);
}
int main() {
    printf("%d", foo(15, 3));
    return 0;
}
```

What is the terminal output returned by ***foo(15, 3)***?

Question 4 (ISRO)

What does the following recursive function return when evaluated with ***mystery(30, 42)***?

```
int mystery(int a, int b) {
    if (b == 0) return a;
    return mystery(b, a % b);
}
```

Question 5 (BARC)

Consider the recursive execution of the function below:

```
void solve(int n) {
    if (n <= 0) return;
    printf("%d ", n);
    solve(n - 1);
    solve(n - 2);
}
```

What is the total number of times the character stream gets printed when ***solve(4)*** is executed?

Question 6 (NIELIT)

Predict the output of the following C code segment:

```
int run(int n) {
    if (n <= 1) return 1;
    if (n % 2 == 0) return n + run(n - 1);
    return n * run(n - 2);
}
int main() {
    printf("%d", run(5));
}
```

Question 7 (GATE)

Consider the recursive function segment:

```
int bar(int n) {
    if (n == 1) return 0;
    else return 1 + bar(n / 2);
}
int foo(int n) {
    if (n == 1) return 1;
    else return 1 + foo(bar(n));
}
```

What is the smallest positive integer n for which ***foo(n)*** returns 5?

Question 8 (DRDO)

Predict the sequence printed by the program call ***rec(3)***:

```
void rec(int x) {
    if (x > 0) {
        rec(--x);
        printf("%d ", x);
        rec(x--);
    }
}
```

Question 9 (GATE)

Consider the following functional representation:

```
int G(int n) {
    if (n <= 0) return 0;
    if (n == 1) return 1;
    return G(n - 1) + G(n - 2);
}
```

How many activation records (stack frames) are created in total to compute ***G(6)***?

Question 10 (VIZAG STEEL PSU)

What will the following program output?

```
int check(int x) {
    static int k = 5;
    if (x == 0) return k;
    k += x;
    return check(x - 1);
}
int main() {
    printf("%d", check(3));
}
```

SECTION B: ARRAYS & POINTER ARITHMETIC

Question 11 (GATE CS 2025)

Consider the following fragment of C code:

```
void modify_array(int *arr, int size) {
    for(int i = 0; i < size - 1; i++) {
        *(arr + i + 1) += *(arr + i);
    }
}
int main() {
    int data[] = {2, 3, 4, 5};
    modify_array(data, 4);
    printf("%d", data[3]);
}
```

What is the exact value output by ***data[3]***?

Question 12 (GATE CS 2020)

Consider the following multi-dimensional initialization in C:

```
int main() {
    int a[3][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
    int *p = a[0] + 1;
    printf("%d", *(p + 5));
}
```

What is the terminal integer output generated?

Question 13 (GATE CS 2022)

Consider a two-dimensional integer array $M[10][10]$ stored in row-major order with a base address of 1000. If each integer occupies 4 bytes of memory, what is the exact physical memory address of element $M[5][4]$?

Question 14 (ISRO)

An array $A[-5...5][2...10]$ is stored in column-major order. If the base address is 2000 and each element requires 2 bytes, find the physical address of element $A[0][5]$.

Question 15 (BARC)

What is printed by the following pointer arithmetic program block?

```
int main() {
    char *cities[] = {"Delhi", "Mumbai", "Kolkata", "Chennai"};
    char **ptr = cities;
    ptr += 2;
    printf("%s", *ptr + 2);
}
```

Question 16 (NIELIT)

Given the array initialization $int\ arr[5] = \{10, 20, 30, 40, 50\};$, determine the programmatic evaluation of the expression $*(\&arr[1] + 2)$.

Question 17 (GATE)

Consider the code segment below:

```
int main() {
    int a[] = {10, 20, 30, 40, 50, 60};
    int *p[] = {a, a + 1, a + 2, a + 3};
    int **pp = p;
    pp++;
    printf("%d %d", pp - p, **pp);
}
```

Question 18 (DRDO)

What is the size (in bytes) allocated on a 64-bit compiler execution system for the following definition?

```
int (*ptr)[15];
```

Question 19 (GATE)

An array implementation variable is defined as ***double grid[15][25][10];***. If the base location starts at structural address ***0***, how many double elements must be stepped past to read the component item ***grid[5][10][2]*** under row-major criteria?

Question 20 (PSU EXAM)

Predict the console output of the execution loop:

```
int main() {
    int numbers[5] = {1, 2, 3, 4, 5};
    int *p = numbers;
    printf("%d ", *(p++));
    printf("%d", *++p);
}
```

SECTION C: FUNCTIONS, SCOPE & PARAMETER PASSING

Question 21 (GATE CS 2024)

Consider the execution of the following structural function logic:

```
int process(int *p, int *q) {
    *p = *p + *q;
    *q = *p - *q;
    *p = *p - *q;
    return *p;
}
int main() {
    int a = 12, b = 15;
    int res = process(&a, &b);
    printf("%d %d %d", a, b, res);
}
```

Question 22 (GATE CS 2019)

Consider the static functional block:

```
int counter(int i) {
    static int count = 0;
    count = count + i;
    return count;
}
```

If evaluated left-to-right sequentially, what is the output computed by the numerical expression ***counter(10)*** + ***counter(20)*** + ***counter(30)***?

Question 23 (ISRO)

What parameter passing style does the following code assume if the output printed inside main is *25*?

```
void calc(int x) {
    x = x * x;
}
int main() {
    int val = 5;
    calc(val);
    printf("%d", val);
}
```

Question 24 (BARC)

Consider the following C function containing scope indicators:

```
int x = 10;
void print_val() {
    printf("%d ", x);
}
void execute() {
    int x = 20;
    print_val();
}
int main() {
    execute();
}
```

What is printed if the compiler applies lexical (static) scoping rules versus dynamic scoping rules?

Question 25 (NIELIT)

Determine the exact functional value printed by the following program:

```
void step(int *val) {
    static int modifier = 2;
    *val += modifier++;
}
int main() {
    int num = 10;
    step(&num);
    step(&num);
    printf("%d", num);
}
```

Question 26 (GATE)

What will the output sequence be of the execution given below?

```
void swing(int *p, int *q) {
    p = q;
    *p = 20;
}
int main() {
    int a = 5, b = 10;
    swing(&a, &b);
    printf("%d %d", a, b);
}
```

Question 27 (DRDO)

What does the function definition ***void (*f)(int, int*)*** signify in standard modular C architecture declarations?

Question 28 (GATE)

Evaluate the printed output sequence of the block below:

```
int swap_eval(int x, int y) {
    int temp = x; x = y; y = temp;
    return temp;
}
int main() {
    int i = 3, j = 8;
    j = swap_eval(i, swap_eval(j, i));
    printf("%d %d", i, j);
}
```

Question 29 (PSU EXAM)

How many variables are active inside the active scope duration execution window of the caller function framework when executing statements inside local function nested blocks?

Question 30 (GATE)

Determine what values the following block prints out:

```
void modify(int a, int *b) {
    a = *b * 2;
    *b = a + 1;
}
int main() {
    int x = 3, y = 5;
    modify(x, &y);
    printf("%d %d", x, y);
}
```

SECTION D: ADVANCED MIXED STRUCTURES

Question 31 (GATE CS 2024)

What is the worst-case number of comparison steps performed by a standard implementation of a recursive binary search variant executed over a sorted structural linear single array vector holding exactly n items?

Question 32 (GATE CS 2025)

Let A be an array containing distinct integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. What is the distance of the array $[2, 5, 3, 1, 4, 2, 6]$?

Question 33 (ISRO)

Trace the following program structural print statement:

```
void trace_arr(int *a, int size) {
    if (size <= 0) return;
    trace_arr(a + 1, size - 1);
    printf("%d ", a[0]);
}
int main() {
    int arr[] = {1, 3, 5, 7};
    trace_arr(arr, 4);
}
```

Question 34 (BARC)

Predict what value is generated by calling $compute\_sum(arr, 5)$:

```
int compute_sum(int *a, int n) {
    if (n <= 0) return 0;
    return a[n-1] + compute_sum(a, n-1);
}
```

Question 35 (NIELIT)

What does the following functional lookup process achieve over an arbitrary array E?

```
int lookup(int E[], int size) {
    int Y = 0, Z = 0;
    for (int i = 0; i < size; i++) Y = Y + E[i];
    for (int i = 0; i < size; i++) {
        for (int j = i; j < size; j++) {
            Z = 0;
            for (int k = i; k <= j; k++) Z = Z + E[k];
            if (Z > Y) Y = Z;
        }
    }
    return Y;
}
```

Question 36 (GATE)

Consider the following C function:

```
int check_matrix(int matrix[4][4], int r, int c) {
    if (r >= 4 || c >= 4) return 0;
    return matrix[r][c] + check_matrix(matrix, r + 1, c + 1);
}
```

If a standard Identity Matrix *I_{4 imes 4}* is passed into *check_matrix(I, 0, 0)*, what value is returned?

Question 37 (DRDO)

What sequence does the following array-printing recursive block emit?

```
void print_rev(char *str) {
    if (*str == '') return;
    print_rev(str + 1);
    printf("%c", *str);
}
int main() {
    print_rev("GATE2026");
}
```

Question 38 (GATE)

What does the function *f* return when called with *f(arr, 0, 4)* where *arr* = *{12, 7, 30, 15, 4}*?

```
int f(int a[], int i, int j) {
    if (i == j) return a[i];
    int mid = (i + j) / 2;
    int left = f(a, i, mid);
    int right = f(a, mid + 1, j);
    return (left < right) ? left : right;
}
```

Question 39 (PSU EXAM)

Predict the terminal execution footprint of the array logic:

```
int main() {
    int arr[3] = {5, 10, 15};
    int *p = arr;
    printf("%d ", *(p + 1));
    printf("%d", p[2]);
}
```

Question 40 (GATE)

Consider the nested recursive pointer code trace:

```
void recurse_ptr(int **p) {
    if (**p == 0) return;
    **p = **p - 1;
    recurse_ptr(p);
}
int main() {
    int val = 3;
    int *ptr = &val;
    recurse_ptr(&ptr);
    printf("%d", val);
}
```

SECTION E: CONCEPTUAL STRUCTURES & ALGORITHMIC PATTERNS

Question 41 (GATE)

An array A of length n with distinct elements is said to be bitonic if there is an index $1 \le i \le n$ such that $A[1 \dots i]$ is sorted in non-decreasing order and $A[i+1 \dots n]$ is sorted in non-increasing order. What is the best possible asymptotic bound for the worst-case number of comparisons required to search for an element in a bitonic array?

Question 42 (ISRO)

What is the auxiliary space complexity required to run an in-place array inversion loop using a traditional iterative head-to-tail two-pointer swap strategy?

Question 43 (BARC)

Which internal system memory structure manages activation frames when deep nested levels of functional recursive calls occur inside standard run-time compiled C environments?

Question 44 (NIELIT)

Will recursion operate properly in an experimental software compilation environment that utilizes static memory allocation layout assignments for all regional, global, and individual localized variable states? Explain your stance.

Question 45 (GATE)

Consider the following code segment:

```
void f(int *p, int *q) {
    p = q;
    *p = 2;
}
int main() {
    int i = 0, j = 1;
    f(&i, &j);
    printf("%d %d", i, j);
}
```

Question 46 (DRDO)

How many data point comparisons are performed by a bubble sort routing tracking function executing sequentially across an input array vector initialized with elements sorted in descending reverse sequence ordering?

Question 47 (GATE)

Given ***int matrix[5][5];*** allocated sequentially at structural start block point ***2000***, what analytical pointer equation maps the programmatic row-major cell location index notation matching ****(*(matrix + i) + j)***?

Question 48 (PSU EXAM)

What type of structural runtime logic fault error surfaces if an executing thread triggers a recursive function tracking no structural breakout base criterion conditions?

Question 49 (GATE)

Evaluate what the print execution tracks:

```
int fun_block(int *a, int n) {
    if (n <= 0) return 0;
    if (*a % 2 == 0) return *a + fun_block(a + 1, n - 1);
    else return fun_block(a + 1, n - 1);
}
int main() {
    int arr[] = {1, 2, 3, 4, 5, 6};
    printf("%d", fun_block(arr, 6));
}
```

Question 50 (GATE)

Consider the multi-pointer expression tracking problem:

```
int main() {
    char *arr[] = {"alpha", "beta", "gamma"};
    char **ptr[] = {arr, arr + 1, arr + 2};
    char ***p = ptr;
    printf("%s", **++p);
}
```